

Onset and Threefold Classification of Outcomes of Spontaneous Splitting in N-Body Relational-Wave Closed Systems

Unfreezing the generator alone makes the parent state decay exponentially at a rate independent of the seed amplitude, and the outcomes divide into expansion, bounded mixing, and recurrence — introducing the scale-anonymity axiom, with the stopping of splitting as a connection problem

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Abstract

Theorem 5 of Paper 4 [3] established that under a fixed generator neither closure quantities nor energy-like quantities move between rotation planes — a no-go for spontaneous splitting of the state (Consequences 16.5 and 16.6 [1]). Taking this no-go as the starting boundary, this paper answers the following question.

Can multiple waves split spontaneously under the closure condition? If they split, can the subsequent outcomes be classified?

To pose the question precisely, we first separate two layers. The closure $\sum_k x_k^2 = 0$ imposes no constraint on the number of terms $\mathcal{N}$ of a decomposition — any term X_P^2 can always be rewritten as $Y_1^2 + Y_2^2$ over the complex numbers (always solvable, with a continuum of solutions), and no obstacle to $\mathcal{N} \rightarrow \mathcal{N} + 1$ exists on the axiom side. The number of waves is not determined inside the system, and **the layer of existence has no ceiling ($\mathcal{N}$ does not saturate)**. We separate the symbols: $\mathcal{N}$ is the number of terms of a decomposition (layer of existence), and N is the number of bodies, the model parameter of this paper (a structural specification of the representation; the flat representation has M relational waves). What this paper measures is therefore not “permission to split” in the layer of existence. It measures the dynamics of the **readout layer** — register allocation, active planes, freezing. The structures that look finite across the series (the $O(N)$ cap on generator rank [2], the saturation of spatial directions at three [2], register freezing [3], and the outcome classification of this paper) are all saturations of the readout, not finiteness of existence.

Three preparations are made. First, we introduce scale anonymity as Axiom 0.5. The relations, update rules, and readouts of the system may reference no absolute scale, and every construction must be covariant under $Z \rightarrow \lambda Z$ ($\lambda \in \mathbb{C}^*$). This is an unprovable formation rule of the same rank as Axiom 0 (anonymity), and the zero right-hand side of the quadratic closure of Axiom 1 is the unique choice compatible with this axiom within the family $\sum_n x_n^2 = c$. Second, we show that every smooth dynamics preserving the closure sphere is exhausted by the form $\dot{X} = K(X)X$ (K skew-symmetric) (Theorem 2). The fixed generator is merely the special case $K(X) = K_0$, and the search space for “the minimal extension enabling splitting” is thereby closed. Third, as the minimal extension intrinsic to the model, the phase-difference sine generator $K_{ef} = A_{ef} \sin(\theta_f - \theta_e)$ is reconstructed at every step from the current phases $\arg Z(\tau)$ (unfreezing). Being a function of phase differences only, this generator is zeroth-order homogeneous and satisfies Axiom 0.5 without adjustment.

Spontaneity is defined operationally: the parent state must be dynamically unstable, i.e., arbitrarily small perturbations must grow exponentially at an intrinsic rate independent of the perturbation amplitude. Two structures were fixed in the construction of the parent state. Real-coefficient in-plane states have collinear component phases, on which the sine generator vanishes exactly (Theorem 4), so a self-consistent parent must be the circularly polarized eigenmode $v = (p + iq)/\sqrt{2}$; and since an eigenmode advances by a common phase rotation only, it is an exact periodic orbit (Proposition 1; numerical verification: generator variation 9.2×10^{-14} over one period). As a numerical insight, the generator of the self-consistent parent has rank 2 (one rotation plane), and the remaining $M - 2$ dimensions form a static kernel.

In the numerical experiments ($N = 4$, $M = 6$, seed amplitudes $\delta = 10^{-2}$ – 10^{-5} , 6 trials each, 4320 steps; the dormant seed is constructed from a real orthonormal pair in the kernel as $g = (u + iw)/\sqrt{2}$ with $g^\top g = 0$ exact, and the initial state satisfies zero closure to $|Z^\top Z| \leq 3.2 \times 10^{-13}$), all three criteria of

spontaneity held. The initial exponential growth rate of the dormant fraction $f(\tau)$ was invariant over three decades of δ , with medians 3.78×10^{-2} – 3.80×10^{-2} /step; absolute-threshold crossing times were linear in $\log(1/\delta)$ (about 84 steps per decade); and relative-threshold crossing times were δ -independent (about 84 steps) (the growth is not single-exponential: the initial-fit rate and the effective rate derived from crossing times differ; Section 8.2). Decisive is the $\delta = 0$ verification. The applied perturbation is zero, and under exact arithmetic the orbit would be invariant. Nevertheless, the nonzero numerical perturbation of floating-point rounding ($f \sim 10^{-33}$) was amplified by about 20 orders of magnitude, reaching $f > 10^{-10}$ at $\tau \approx 850$ and saturating — strong evidence of transverse instability. Throughout, the real and imaginary parts of the complex bilinear closure (corresponding to the exact splitting conditions of Paper 4) were conserved to deviations $\leq 1.8 \times 10^{-14}$. In the control experiment (same initial conditions, fixed generator), reference-register variation was $\leq 4.5 \times 10^{-11}$: the numerical reconfirmation of the no-go and the onset of splitting were established on the same code. Scale anonymity was verified numerically as well: under the scale transformation $\lambda = 2^{10}$ all ratio observables agreed bitwise, and for generic λ and common phase rotations deviations stayed $\leq 1.1 \times 10^{-13}$. The onset law is also robust in the number of bodies. Runs with the low-rank vertex-decomposition engine (an algorithmic use of the decomposition $K = WJW^T$ of Paper 3) at $N = 40, 300, 1000$ ($M = 780, 44850, 499500$), measuring the dormant fraction without cancellation via projection onto the orthogonal complement of the parent plane, directly observed constant-rate exponential amplification over more than 20 decades from seed depths $f(0) \approx \delta^2 = 10^{-30}$ (Experiment O6, Section 8.8). The rate depends only weakly on N ($3.6 \times 10^{-2} \rightarrow 2.1 \times 10^{-2}$ /step, decreasing monotonically with N within the same order).

In the outcome-classification experiments ($N = 3, 4, 5, 6$, two families of initial conditions — near-parent and generic zero-closure states — 6 trials each, 20000 steps), the outcomes divided into three classes: expansion (a new rotation plane establishes itself and the allocation spreads over all relational waves), bounded mixing (the allocation spreads but no second plane establishes itself), and recurrence (repeated deep returns toward the parent plane). Class selection has a sharp N phase structure. At $N = 3$ expansion is structurally impossible (a skew-symmetric matrix with $M = 3$ has rank ≤ 2); at $N = 4$ zero-closure states did not expand either (both families bounded mixing); for $N \geq 5$ every initial condition examined (2 families $\times$ 6 seeds) expanded — the boundary lies between $N = 4$ and $N = 5$. Furthermore, a controlled contrast experiment (Experiment O5: everything identical except the initial value of the complex bilinear closure) confirmed that the ratio $c = |Z^T Z|/\|Z\|^2$, exactly conserved by the dynamics, stratifies the outcome classes at small N — $N = 4$: bounded mixing at $c = 0$ (3/3) but expansion at $c \approx 0.6$ (3/3); $N = 3$: bounded mixing at $c = 0$ (3/3) but recurrence at $c \approx 0.55$ (3/3). Closure is not only a conservation law; its conserved value selects the outcome (Section 8.7). In addition, in the $N = 4$ parent family the second singular direction σ_2 decayed exponentially without external intervention ($6 \times 10^{-4} \rightarrow 10^{-5}$ over 20000 steps), an observed route by which the system spontaneously refreezes into a single-plane self-consistent configuration. The stopping mechanism of splitting, together with this observation, is the central subject of the sequel.

0. Conclusions

Closure-preserving dynamics is exhausted by $\dot{X} = K(X)X$ (K skew-symmetric); the fixed generator is a special case. Unfreezing the generator alone makes the parent state (circularly polarized eigenmode) decay exponentially without external forcing, while the real and imaginary parts of the closure are conserved to machine precision and the allocation moves spontaneously into dormant directions. The outcomes divide into three classes — expansion, bounded mixing, recurrence — selected by N (structural room) and the conserved ratio $c = |Z^T Z|/\|Z\|^2$.

Spontaneity holds as dynamical instability. The growth rate is invariant over three decades of the seed amplitude δ , and even the exact periodic orbit at $\delta = 0$ collapses from rounding noise alone. Under a deterministic update rule, which direction the state collapses into is decided by microscopic differences not contained in the macroscopic description — determinism and predictability separate in this minimal model.

The outcome classes have an N phase structure. At $N = 3$ expansion is structurally impossible; at $N = 4$ zero-closure states do not expand; for $N \geq 5$ every initial condition examined expanded — the boundary

lies between $N = 4$ and $N = 5$. The controlled contrast (Experiment O5) confirmed that what acts is the exactly conserved ratio $c = |Z^\top Z|/\|Z\|^2$: at $c = 0$ (the zero-closure stratum = the stratum designated by Axiom 1) expansion is suppressed for $N \leq 4$, while at $c \approx 0.5\text{--}0.6$, $N = 4$ expands and $N = 3$ recurs. The value of the conserved quantity stratifies the outcome. In the $N = 4$ parent family the second singular direction decays exponentially of its own accord, heading toward refreezing — evidence that stopping can occur without an external freezing operation, and the central object of the sequel (stopping of splitting).

All statements of this paper belong to the readout layer. The layer of existence ($\mathcal{N}$ itself) is unconstrained by the axioms and does not saturate — every structure that looks finite is a saturation of the readout (Section 1.2).

Scale anonymity (Axiom 0.5) runs through the entire construction. The unfrozen generator is zeroth-order homogeneous as a function of phase differences only; all readouts are ratio quantities; all activity thresholds are relative. Numerically, all ratio observables agreed bitwise under a 2^{10} -fold scale change.

1. Research Question

1.1 The no-go as the starting boundary

Paper 3 [2] proved that the fixed generator of N-body complete pairwise relational waves decomposes into a direct sum of rotation planes and a null space, with the number of rotation modes growing at most linearly. Paper 4 [3] established that each plane carries kinematics isomorphic to the two-body AB system, and its Theorem 5 gave a strong no-go: under a fixed generator, neither the closure quantities Q_j nor the energy-like quantities H_j move between planes or with the kernel (Consequences 16.5 and 16.6 [1]).

Spontaneous splitting of the state — reallocation between registers, activation of new rotation modes — therefore cannot occur in the fixed-generator dynamics as a matter of principle. To ask about splitting, a relational update rule beyond the fixed generator is required. This paper identifies the minimal such extension, determines numerically whether splitting begins, and classifies the outcomes after onset.

This paper is the first of a three-part sequence. First question (this paper): can splitting begin? Second question (sequel): can splitting stop? Third question (sequel): can the conserved quantities generated by splitting differentiate?

1.2 Layer of existence and layer of readout — what this paper measures

The closure $\sum_k x_k^2 = 0$ does not constrain the number of terms $\mathcal{N}$ of a decomposition. The splitting $X_P^2 = Y_1^2 + Y_2^2$ is always solvable over the complex numbers, with a continuum of solutions — the splitting map is not a “hard-to-construct missing piece” but an **underdetermined freedom**. In the layer of existence the number of waves is arbitrary (it may be infinite) and does not saturate.

Distinction of symbols: $\mathcal{N}$ (number of terms of a decomposition, layer of existence) and N (number of bodies, structural specification of the model = external input) are different quantities. The N used by the experiments of this paper for parameterization is the latter, and the number of terms of its flat representation is the number of relational waves $M = \binom{N}{2}$.

By contrast, every finiteness the series has established belongs to the readout layer.

Quantity	Behavior	Source
Number of relational waves M	$O(N^2)$, unbounded	Definition
Number of terms of a decomposition $\mathcal{N}$ (layer of existence)	Does not saturate (the axioms admit any $\mathcal{N}$)	Axiom 1
Number of bodies N (model specification)	External input (structural specification of the representation)	Section 4.1
Generator rank	Capped at $\leq 2N$	Paper 3 [2]
Uniquely readable spatial directions	Saturates at 3	Paper 3 [2]

Quantity	Behavior	Source
Readout registers	Frozen under a fixed generator (no-go)	Paper 4 [3]

Existence opens without limit; only the readout saturates

A world that looks finite is a consequence of readout saturation, not of the finiteness of existence. All “splitting” in this paper is a statement of the readout layer — register reallocation and the waxing and waning of active planes — and the operation of increasing the term count $\mathcal{N}$ is not an event requiring dynamical justification (see limitation (i) in Section 9).

1.3 Operational definition of spontaneity

Applying a splitting map by hand proves no spontaneity. In this paper spontaneity is defined operationally as dynamical instability.

1. The growth rate of the dormant component does not depend on the seed amplitude δ (exponential growth at an intrinsic rate).
2. Absolute-threshold crossing times are linear in $\log(1/\delta)$; relative-threshold crossing times are δ -independent.
3. The closure is conserved exactly throughout (the splitting is not caused by a breaking of closure).

As the strongest verification we place $\delta = 0$. The applied perturbation is zero; under exact arithmetic the orbit is invariant. If the only seed is the nonzero numerical perturbation of floating-point rounding, and that seed is amplified into collapse, this is strong evidence of transverse instability — the parent state collapses under arbitrarily small perturbations.

1.4 What this paper decides

1. How is the totality of closure-preserving dynamics characterized? Where does the fixed generator sit inside it?
2. Is the minimal intrinsic extension (unfreezing) consistent with Axiom 0.5?
3. Does a self-consistent parent state (an exact periodic orbit of the unfrozen dynamics) exist? What is its structure?
4. Is the parent state unstable? Do the three criteria of spontaneity hold?
5. Can the outcomes after onset be classified? What decides class selection?
6. Is all of the above scale-covariant?

No background space, distance matrix, or external observer is introduced. No cosmological or field-theoretic physical names are used in this paper (Axiom 0).

2. Classification of Claims

Subject	Classification	Position in this paper
N-body complete pairwise relational-wave model	Model definition	Implementation of Axiom 16 [1]; inherited from [2,3]
Non-constraint of $\mathcal{N}$ (the term count is not internally determined and does not saturate)	Consequence of Axiom 1	Section 1.2. No ceiling in the layer of existence
The measurement target of this paper is the readout layer (not the layer of existence)	Explicit framework	Section 1.2. All finiteness is readout saturation
Scale anonymity	Axiom (first appearance)	Axiom 0.5. Unprovable formation rule. Section 3

Subject	Classification	Position in this paper
The choice of zero on the right-hand side of Axiom 1	Consequence of Axiom 0.5	The unique scale-free quadratic closure. Section 3
Exact conservation of the complex square sum under orthogonal updates	Mathematical consequence	Theorem 1
Exhaustion of closure-preserving dynamics by $\dot{X} = K(X)X$	Mathematical consequence	Theorem 2. Closure of the search space
Zeroth-order homogeneity of the unfrozen sine generator	Mathematical consequence	Theorem 3. First application of Axiom 0.5
Vanishing of the generator on collinear-phase states	Mathematical consequence	Theorem 4. The parent must be circularly polarized
Exact periodicity of the eigenmode parent	Mathematical consequence	Proposition 1
Effective rank 2 of the self-consistent parent's generator ($N = 4$)	Numerical insight	Unproven. Section 5. Multi-plane for larger N (§5.3)
Exponential instability of the parent (rate independent of δ)	Numerical confirmation	Experiment O1. Section 8
Collapse from rounding noise at $\delta = 0$	Numerical confirmation	Experiment O1. Section 8
No-go under the fixed generator	Derived consequence [1,3]	Numerically reconfirmed in control Experiment N
Threefold outcome classification (expansion, bounded mixing, recurrence)	Numerical classification	Experiment O4. Classification thresholds preliminary
Impossibility of expansion at $N = 3$	Derived consequence	A skew-symmetric matrix with $M = 3$ has rank ≤ 2
N phase structure ($N \leq 4$ no expansion; $N \geq 5$ expansion in all trials)	Numerical confirmation	Experiment O4
Outcome stratification by the conserved ratio $c = Z^T Z /\ Z\ ^2$ (expansion suppressed for $N \leq 4$ at $c = 0$)	Numerical confirmation	Experiment O5 (controlled contrast). Section 8.7
Persistence of the onset law at large N ($N = 40$ – 1000 , seed depth $f(0) \approx 10^{-30}$, constant-rate amplification over 20+ decades)	Numerical confirmation	Experiment O6. Section 8.8
Spontaneous exponential decay of σ_2 (refreezing candidate)	Numerical observation / working hypothesis	Confirmation of refreezing requires additional criteria such as $\ K(\tau+1) - K(\tau)\ $. Sequel
Origin of the saturation ceiling $f \approx 0.335$ – 0.338	Unresolved	No identification with any specific constant is made
Relation to modulational instability [5,6]	Structural similarity	Isomorphic phenomenology, different coupling structure (Section 9)
Recurrence of splitting and recombination	Numerical confirmation	Phenomenology isomorphic to FPUT-type recurrence [7,8]
An explicit splitting map increasing the component count	Not constructed	Measured here as growth of the active-plane count. Connection problem
Stopping mechanism of splitting	Connection problem	Sequel (second question)

3. Axiom 0.5: Scale Anonymity

3.1 Full statement of the axiom

Axiom 0.5 (scale anonymity) The relations, update rules, and readouts of the system may reference no absolute scale. Every construction must be covariant under $Z \rightarrow \lambda Z$ ($\lambda \in \mathbb{C}^*$): the relations (closure conditions) are homogeneous, the generator is zeroth-order homogeneous ($K(\lambda Z) = K(Z)$), the state map is first-order homogeneous, and the readouts are ratio quantities only. No means exists to measure an absolute scale from inside the system.

3.2 Position

Axiom 0 is the formation rule “no physical names may be referenced.” Axiom 0.5 is the coequal formation rule “no absolute scale may be referenced,” placed immediately after Axiom 0 and before Axiom 1. The two are two clauses of a single idea: no names, no scale — only relations and ratios can be read.

This axiom is not a derived consequence. It is unprovable in the sense that all readouts are ratios and no means of measuring an absolute scale from inside the system exists in principle. It has the same rank as Axiom 1 (zero-square-sum closure), which is likewise unprovable.

3.3 The choice of the form of Axiom 1

As a quadratic closure condition, any of $\sum_n x_n^2 = c$ was conceivable; but $c \neq 0$ breaks under $Z \rightarrow \lambda Z$ as $c \rightarrow \lambda^2 c$. Within the family $\sum_n x_n^2 = c$, the only dilation-invariant right-hand side is

$$\sum_n x_n^2 = 0$$

($c = 0$). If Axiom 0.5 is placed first, the zero on the right-hand side of Axiom 1 is fixed with no room for choice within this family. If general quadratic forms $Z^T G Z = 0$ are admitted, other homogeneous closures can be written, so the claim is not “the quadratic closure itself is unique” but “zero is the unique right-hand side of $\sum x_n^2 = c$.” The choice of the form $\sum x_n^2$ ($G = I$) is required by the name-permutation symmetry (anonymity). Moreover, the compact part $|\lambda| = 1$ of the covariance group $\mathbb{C}^*$ is the common-phase invariance of the zero-square-sum closure: the unreadability of phase and the unreadability of scale are the argument part and the modulus part of one and the same group.

3.4 Consequences within this paper

Axiom 0.5 performs three concrete jobs in this paper. First, it cuts the search space of extended update rules: the generator must be zeroth-order homogeneous and cannot depend on absolute amplitudes (phases and amplitude ratios only). Second, it restricts all readout thresholds to relative values: the activity threshold for planes, the seed amplitude, and the recurrence criterion are all defined as ratios. Third, it provides the acceptance criterion for numerical verification: the ratio observables of a scale-transformed orbit must agree exactly (Section 8.4).

After publication of this paper, this axiom will be incorporated into layer 0 in the next version (v8) of the basic axiom system; Axiom 0 and Axiom 0+ will then be merged as clauses of a single Axiom 0 (anonymity).

4. Preliminaries: Model and Conservation Structure

4.1 State and generator

Consider the complete pairwise relational waves of N bodies. The relational waves correspond to the edges of the complete graph K_N ; there are $M = \binom{N}{2}$ of them. The state is a complex vector $Z \in \mathbb{C}^M$ with phases $\theta_e = \arg Z_e$. The line-graph adjacency matrix A ($A_{ef} = 1$ when edges e, f share a body) and the phase-difference sine generator

$$K_{ef} = A_{ef} \sin(\theta_f - \theta_e)$$

are inherited from the definitions of Paper 3 [2] (since $\theta_e = \arg Z_e$ is undefined at $Z_e = 0$, we adopt the convention $\arg 0 := 0$; in the initial conditions of this paper all components are nonzero). K is automatically skew-symmetric and is normalized in spectral norm (after normalization $\sigma_{\max} = 1$, which makes all subsequent thresholds relative). The update is the Cayley transform [10]

$$U = (I - \gamma K)^{-1}(I + \gamma K), \quad \gamma = \tan \frac{\pi}{144}.$$

If K is skew-symmetric, U is a real orthogonal matrix.

Fixed generator (control): K is evaluated at the phases $\arg Z(0)$ at $\tau = 0$ and frozen. This is the dynamics of Papers 3 and 4.

Unfreezing (the minimal extension of this paper): K is reconstructed at every step from the current phases $\arg Z(\tau)$. What is introduced is not a new coupling rule but only the release of the freezing of a generator that was defined as a function of phase differences from the start.

4.2 Theorem 1 (exact conservation of the complex square sum under orthogonal updates)

An update $Z' = UZ$ by a real orthogonal matrix U conserves the complex bilinear square sum and the Hermitian norm simultaneously and exactly:

$$Z'^T Z' = Z^T U^T U Z = Z^T Z, \quad Z'^\dagger Z' = Z^\dagger Z.$$

Proof Immediate from $U^T U = I$ and the realness of U . ■

Conservation of the complex square sum $Z^T Z$ is the simultaneous conservation of two conditions, its real and imaginary parts. These are the same two conditions demanded by the local splitting condition $X_P^2 = \sum_k Y_k^2$; the dynamics of this paper never breaks this closure at any point of the splitting process.

4.3 Theorem 2 (exhaustion of closure-preserving dynamics)

Every smooth flow on the closure sphere $\{X : X^T X = R^2\}$ ($R^2 > 0$, $X \neq 0$) can be written, with a skew-symmetric matrix-valued function $K(X)$, in the form

$$\dot{X} = K(X)X,$$

and conversely every flow of this form preserves the closure.

Proof Conservation: $\frac{d}{d\tau} X^T X = 2X^T K(X)X = 0$ (skew-symmetry). Exhaustion: for a tangent vector field $V(X)$ on the sphere ($X^T V = 0$), set $K(X) = \frac{VX^T - XV^T}{X^T X}$; then K is skew-symmetric and $K(X)X = V$. ■

Remark (scope) (i) For isotropic states ($X^T X = 0$, $X \neq 0$) the denominator of the above construction vanishes; but taking Y with $Y^T X = 1$ and setting $K = VY^T - YV^T$ gives $KX = V - Y(V^T X) = V$ (using the tangency condition $V^T X = 0$), which repairs the construction. $X = 0$ is excluded. (ii) The theorem exhausts continuous-time flows; the state-dependent Cayley update of this paper is one discretization within this class. No exhaustion of all discrete maps is claimed.

Thus a “state-dependent generator” is not one of many extension candidates but the totality of closure-preserving dynamics. The fixed generator is merely the special case $K(X) = K_0$, and the search for the minimal extension enabling splitting may be conducted inside this space.

4.4 Theorem 3 (scale covariance of the unfrozen generator)

The unfrozen sine generator is zeroth-order homogeneous: for any $\lambda \in \mathbb{C}^*$,

$$K(\lambda Z) = K(Z).$$

Proof Write $\lambda = |\lambda|e^{i\varphi}$. Then $\arg(\lambda Z_e) = \arg Z_e + \varphi$, and the phase differences $\theta_f - \theta_e$ depend on neither $|\lambda|$ nor φ . Spectral-norm normalization is a ratio and hence invariant. ■

The unfrozen dynamics therefore satisfies Axiom 0.5 without adjustment. Since the update is linear (U is a zeroth-order function of Z), the entire orbit commutes with $Z \rightarrow \lambda Z$, and the orbits of ratio observables are exactly scale-independent.

4.5 Two invariants and the conserved ratio c

By Theorem 1 the dynamics conserves two quantities simultaneously and exactly: the Hermitian norm $\|Z\|^2$ and the complex bilinear square sum $Z^\top Z$. Their ratio

$$c = \frac{|Z^\top Z|}{\|Z\|^2} \in [0, 1]$$

is a conserved dimensionless ratio and a readable quantity conforming to Axiom 0.5 (the upper bound $c \leq 1$ follows from the triangle inequality $|\sum Z_e^2| \leq \sum |Z_e|^2$; equality is equivalent to squared-phase alignment of all components). c stratifies the orbits: states with different c cannot be carried into one another by the dynamics.

- $c = 0$ **stratum**: zero closure $Z^\top Z = 0$. The stratum designated by Axiom 1; Experiments O1 and O4 of this paper run here
- $c > 0$ **strata**: the closure convention $X^\top X = R^2 > 0$ of Papers 3 and 4 [2,3] corresponds here. The theorems of both papers are linear algebra and do not depend on c , but the standard initial states of their experiments (imaginary seed amplitude $s = 0.35$) run in the intermediate stratum $c = R^2/(R^2 + 2s^2) \approx 0.80$. Purely real states (for which $X^\top X = \|X\|^2$ holds identically) are the limiting case $c = 1$
- Generic complex states take intermediate c

That the series has moved back and forth between “real sphere” and “zero closure” formulations is not a contradiction but a use of strata labeled by the conserved quantity c . And as Section 8.7 shows, at small N the value of c actually changes the outcome class — the distinction of strata is physics, not representational convenience.

5. The Parent State: Necessity of the Circularly Polarized Eigenmode

5.1 Theorem 4 (vanishing of the generator on collinear-phase states)

On a state whose component phases are collinear ($\theta_e \in \{\psi, \psi + \pi\}$ for some ψ), the sine generator is exactly zero.

Proof For any e, f we have $\theta_f - \theta_e \in \{0, \pm\pi\}$, hence $\sin(\theta_f - \theta_e) = 0$. ■

A state combining plane basis vectors with real coefficients, $Z = e^{i\psi}(\cos \varphi p + \sin \varphi q)$ (p, q real vectors), falls under this condition since every component is a real multiple of $e^{i\psi}$. That is, linearly polarized in-plane states degenerate the generator and cannot carry self-consistent dynamics. **The exact parent constructed in this paper is the circularly polarized eigenmode.** What Theorem 4 excludes is collinear-phase states; Proposition 1 gives a sufficient condition for a circularly polarized eigenmode to be an exact periodic parent. That no other kind of non-collinear self-consistent periodic orbit exists is not shown (open).

5.2 Proposition 1 (exact periodicity of the eigenmode parent)

Let (σ, p, q) be the maximal singular-value plane of K (orientation convention $Kp = +\sigma q$) and

$$v = \frac{p + iq}{\sqrt{2}}, \quad Kv = -i\sigma v$$

its eigenmode. If v is self-consistent with its own generator ($K(\arg v) = K$), then under the unfrozen dynamics v is an exact periodic orbit.

Proof $Uv = \frac{1 - i\gamma\sigma}{1 + i\gamma\sigma} v = e^{-i\theta} v$ ($\theta = 2 \arctan \gamma\sigma$). The update is a common phase rotation; the phases of all components shift uniformly, so the phase differences are invariant. Hence $K(\arg Uv) = K(\arg v)$ is preserved, and the same rotation repeats at every subsequent step. ■

The self-consistent eigenmode was constructed numerically by the fixed-point iteration $\theta \leftarrow \arg v(K(\theta))$. Since the in-plane $SO(2)$ gauge acts only on the common phase of v (rotating p, q by an angle α sends $v \rightarrow e^{-i\alpha} v$), the iteration does not depend on the gauge choice. The reached parent had a median eigenmode-equation residual of 8.9×10^{-13} ; in the $\delta = 0$ verification, over one period (144 steps) the generator variation stayed at 9.2×10^{-14} and the dormant fraction remained exactly zero.

5.3 Numerical insight: the parent’s generator has rank 2

The generator $K(\arg v)$ of the self-consistent parent had effective rank 2 (one rotation plane, $\sigma = 1$) in every configuration examined at $N = 4$, the remaining $M - 2$ dimensions being the kernel (static subspace). The parent state sees only its own plane; all dormant directions belong to the kernel. No proof of this property has been obtained, and it is likely specific to $N = 4$ — separately constructed self-consistent parents at larger N had generators with several planes ($\sigma_2/\sigma_1 \approx 0.35$ – 0.5 ; treated in the sequel). The design of the dormant seed in this paper (kernel directions) is based on this $N = 4$ structure.

The dormant seed is placed so as to preserve zero closure: constructing $g = (u + iw)/\sqrt{2}$ from a real orthonormal pair (u, w) in the kernel gives $g^\top g = 0$ exactly, and together with $v^\top g = 0$ the initial state preserves zero closure. For the $N \geq 5$ parent families of Sections 8.6 and 8.7 the parent generator is multi-plane and the “kernel direction” design does not apply; the seeds of those families are constructed by projection onto the orthogonal complement of the parent plane $\text{span}\{v, \bar{v}\}$ (Experiment O4: complex Gaussian projection; Experiment O5 parentZ0: a zero-closure pair from a real orthonormal pair inside the complement). At $N = 3$ the orthogonal complement of the parent plane is only one-dimensional and no zero-closure perturbation orthogonal to the parent exists (an isotropic vector needs two real dimensions) — in that case a real seed $g = u$ is used and a closure residual of $O(\delta^2)$ is explicitly tolerated (the $N = 3$ parent family of Section 8.6).

In addition, in 4 of 6 parent preparations at $N = 5$ and 1 of 6 at $N = 6$, the eigenmode residual stalled at $\sim 9 \times 10^{-2}$ (worsened convergence of the fixed-point iteration, or possibly nonexistence of the fixed point at larger N). The “parents” of those trials are approximate parents. The conclusions of Section 8.6 are unchanged when restricted to trials with residuals $\sim 10^{-15}$ and are robust, but the existence problem of parents at larger N remains open.

6. Readout Definitions for Splitting

Splitting is read through three ratio quantities (all relative quantities conforming to Axiom 0.5).

Dormant fraction For the parent plane (p_1, q_1) of the reference generator $K(\arg Z(0))$,

$$f(\tau) = 1 - \frac{|p_1^\top Z|^2 + |q_1^\top Z|^2}{Z^\dagger Z}.$$

The share of the allocation outside the parent plane (in the kernel). This is the readout of splitting as register reallocation.

Second singular direction The second singular value σ_2 of the instantaneous generator $K(\arg Z(\tau))$ (after normalization $\sigma_1 = 1$, so σ_2 is a ratio). The relative threshold $\sigma_2 > 0.05$ defines an active plane. This is the readout of the establishment of a new rotation mode — corresponding to an increase of the component count. Note that an absolute threshold cannot be placed by Axiom 0.5; numerical dust $\sigma_2 = O(\delta)$ must not be counted as a plane.

Participation ratio The participation ratio of edge amplitudes

$$\text{PR}(\tau) = \frac{\left(\sum_e |Z_e|^2\right)^2}{\sum_e |Z_e|^4} \in [1, M].$$

The readout of the spread of the allocation [11].

7. Experimental Design

Experiment N (control)

The identical initial condition (parent + dormant seed) is updated with the fixed generator (frozen at $\tau = 0$). By Consequences 16.5 and 16.6 [1,3], the reference plane registers and the kernel allocation should be individually conserved to machine precision. This doubles as a soundness check of the measurement system and a numerical reconfirmation of the no-go.

Experiment O1 (onset of splitting)

The identical initial condition is updated with unfreezing. $N = 4$ ($M = 6$), seed amplitudes $\delta = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$, 6 random seeds each, 4320 steps. The dormant seed is placed from a real orthonormal pair in the kernel as $g = (u + iw)/\sqrt{2}$ ($g^\top g = 0$ exact), and the initial state satisfies zero closure to $|Z^\top Z| \leq 3.2 \times 10^{-13}$. The initial dormant fraction came to $f(0) \approx 0.62 \delta^2$. In addition a $\delta = 0$ verification run (pure eigenmode, no perturbation) is placed.

Scale verification

For a representative condition ($\delta = 10^{-3}$), the ratio observables of orbits with $Z_0 \rightarrow \lambda Z_0$ are compared: $\lambda = 2^{10} = 1024$ (a scale change with exactly zero floating-point rounding error), $\lambda = 987.654$ (generic value), $\lambda = e^{0.7i}$ (common phase rotation). Additionally the orbit-divergence rate against a twin perturbation (one component changed by a relative 10^{-13}) is compared, to separate deviations at generic λ — dynamical amplification of rounding differences — from any breaking of scale symmetry.

Experiment O4 (outcome classification)

$N = 3, 4, 5, 6$ ($M = 3, 6, 10, 15$), two families of initial conditions (parent: self-consistent parent + $\delta = 10^{-3}$ seed; generic: uniformly random complex state), 6 random seeds each, 20000 steps. The classification metrics are the late-window (second-half) minimum of f (recurrence depth), the residence fraction of $\sigma_2 > 0.05$ (establishment of expansion), and the participation ratio. The (preliminary) classification rules are:

- **Expansion:** σ_2 active residence fraction > 0.5
- **Recurrence:** late-window min $f < 0.05$
- **Bounded mixing:** everything else

Experiment O5 (controlled contrast of the closure convention)

With everything else identical (dynamics, classification metrics, 20000 steps), four families contrast only the initial conserved ratio c : genericZ0 (generic state, $Z^\top Z = 0$ exact), genericNZ (generic state, complex Gaussian, $c \approx 0.3$ – 0.6), parentZ0 (parent + zero-closure kernel seed), parentNZ (parent + complex Gaussian seed, $c \lesssim \delta^2$). $N = 3$ – 6 , 3 random seeds each. Whether c changes the outcome class is judged directly.

Experiment O6 (large-N onset runs)

To confirm that the onset law (Experiment O1) is not a peculiarity of small N , onset runs at $N = 40$ ($M = 780$), $N = 300$ ($M = 44850$), and $N = 1000$ ($M = 499500$) are performed with the low-rank vertex-decomposition engine. The engine uses the vertex decomposition $K = WJW^\top$ of Paper 3 as an algorithm, reducing one step to $O(N^3)$ (agreement with the dense implementation verified at $N = 12$: maximum orbit deviation 5.6×10^{-14} over 300 steps).

The seed is $f(0) \approx \delta^2 = 10^{-30}$ in energy ratio ($\delta = 10^{-15}$, near the lower limit at which the seed’s relative amplitude against the parent components approaches the double-precision rounding unit). To measure this depth, the dormant fraction is measured not by the subtraction $1 - h_1/h_{\text{total}}$ (whose cancellation floor is 10^{-16}) but directly by the projection norm onto the orthogonal complement of the parent plane, $f = \|Z - p(p^\top Z) - q(q^\top Z)\|^2/\|Z\|^2$. The parent’s self-consistent iteration is converged to residuals at the

10^{-12} – 10^{-11} level (achieved residuals are listed in the results table; the energy of the construction residual, $\sim 10^{-24}$ – 10^{-23} , sets the measurement floor). The seed is $g = (u + iw)/\sqrt{2}$ ($g^\top g = 0$ exact) from a real orthonormal pair in the kernel (for large N , the orthogonal complement of $\text{span}(W)$). One random seed per N ; runs are tracked to 1500 steps past the crossing of the threshold $f > 0.05$ (cap 9000 steps).

8. Results

8.1 Control (Experiment N): numerical reconfirmation of the no-go

Under the fixed generator, the maximum variation of the reference register ratios was $\leq 4.5 \times 10^{-11}$ across all trials, and the dormant kernel allocation remained at δ^2 without any growth — exactly as Consequences 16.5 and 16.6 predict.

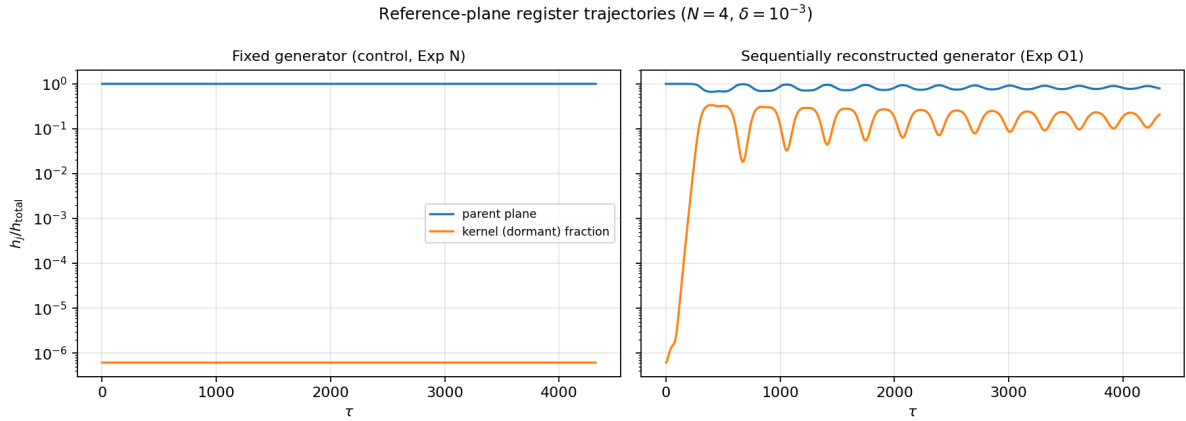


Figure 1: Fixed generator versus unfreezing

Figure 1: Orbits of the reference plane registers ($N = 4$, $\delta = 10^{-3}$). Left: fixed generator (control) — the parent plane and kernel allocations are frozen to machine precision. Right: unfreezing — from the identical initial condition the allocation moves spontaneously.

8.2 Onset of splitting (Experiment O1): the three criteria of spontaneity hold

δ	10^{-2}	10^{-3}	10^{-4}	10^{-5}
initial growth rate (median, /step)	3.795×10^{-2}	3.783×10^{-2}	3.781×10^{-2}	3.781×10^{-2}
$\tau_c(f \geq 10^{-2})$ (median)	137.5	222	306	390
$\tau_c(f \geq 10f(0))$ (median)	83.5	84	84	84

The growth rate is invariant over three decades of δ (the rate depends on the random seed — the parent configuration — but not on δ). Absolute-threshold crossing times are linear in $\log(1/\delta)$ (about 84 steps per decade); relative-threshold crossing times are δ -independent. Criteria 1 and 2 hold.

The growth is not single-exponential, however. Against the crossing interval $2 \ln 10/g \approx 122$ steps predicted from the initial-fit-window rate $g \approx 3.8 \times 10^{-2}$, the measured interval is about 84 steps, so the effective rate defined from crossing times is $2 \ln 10/84 \approx 5.5 \times 10^{-2}$. The initial transient and the main regime have different rates; both values are reported as a limitation of the single-exponential approximation.

Figure 2: Growth of the dormant fraction $f(\tau)$ (median over random seeds). Four parallel exponential ramps shifted by $\log(1/\delta)$ — the classic signature of an instability. After saturation the curves enter a universal ceiling ($f \approx 0.335$ – 0.338) and back-and-forth oscillation.

Figure 3: Threshold crossing times. Absolute thresholds (circles, squares) are linear in $\log_{10}(1/\delta)$; the relative threshold (triangles) is δ -independent.

Figure 4: Distribution of initial exponential growth rates. The point pattern of each δ column is identical — the rate is decided by the random seed (parent configuration), not by the seed amplitude.

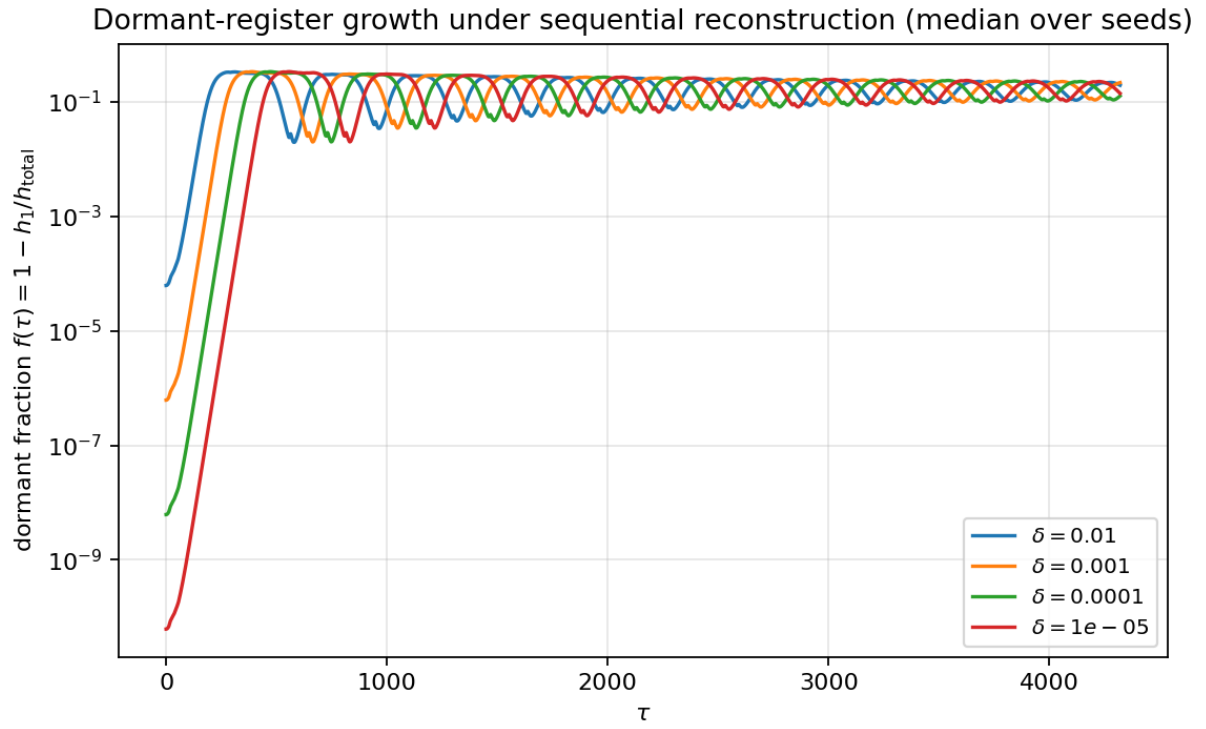


Figure 2: Dormant-fraction growth for the series

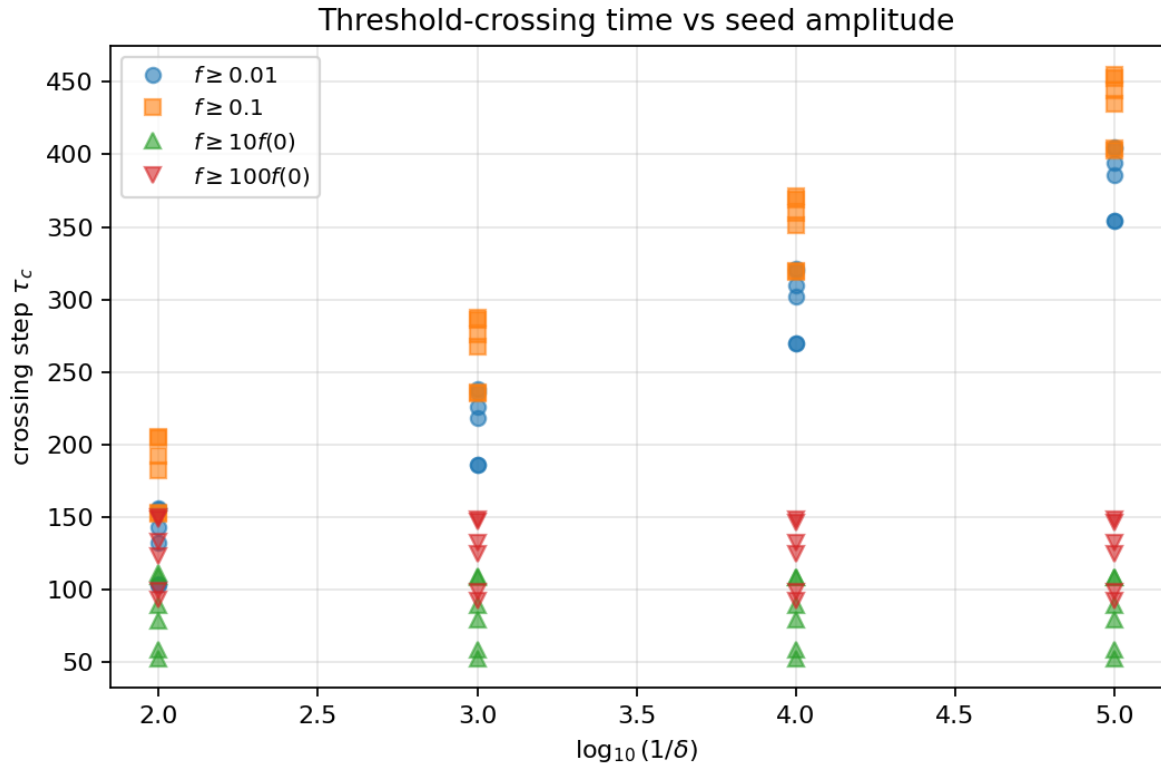


Figure 3: dependence of crossing times

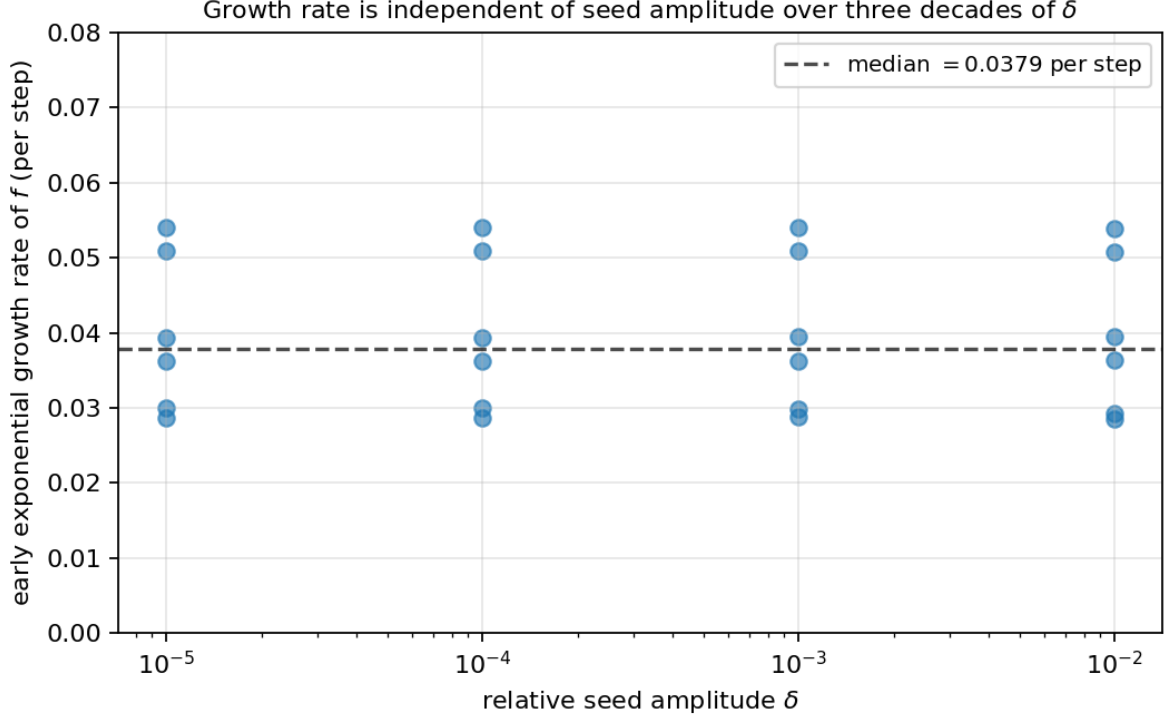


Figure 4: independence of the growth rate

Decisive is $\delta = 0$. The applied perturbation is zero, and under exact arithmetic the orbit would be invariant. Nevertheless the orbit collapsed from the rounding-noise floor ($f \sim 10^{-33}$) through an exponential ramp of about 20 decades, reaching $f > 10^{-10}$ at $\tau \approx 850$ and then saturating to the same ceiling — amplification of the nonzero numerical perturbation of rounding, and strong evidence of transverse instability.

Figure 5: The $\delta = 0$ verification. An orbit that was exactly periodic over one period (generator variation 9.2×10^{-14} , $f = 0$) collapses spontaneously with nothing but floating-point rounding as the seed.

8.3 Exact conservation of the closure

	$ \Delta \text{Re } Z^T Z $	$ \Delta \text{Im } Z^T Z $	$ \Delta Z^\dagger Z /Z^\dagger Z$
fixed generator	$\leq 1.1 \times 10^{-14}$	$\leq 8.3 \times 10^{-15}$	$\leq 1.1 \times 10^{-12}$
unfreezing	$\leq 1.7 \times 10^{-14}$	$\leq 1.2 \times 10^{-14}$	$\leq 1.8 \times 10^{-14}$

The onset of splitting proceeds without breaking the real and imaginary parts of the complex square-sum closure (the two conditions demanded by the local splitting condition). Criterion 3 holds.

8.4 Numerical verification of scale anonymity and separation of sensitivity

transformation	$\max \Delta(\text{ratio observable}) $
$\lambda = 2^{10} = 1024$	0 (bitwise agreement)
$\lambda = 987.654$	2.9×10^{-14}
$\lambda = e^{0.7i}$ (common phase)	1.1×10^{-13}
twin perturbation 10^{-13} (reference)	1.6×10^{-11}
twin perturbation, fixed generator (reference)	7.8×10^{-16}

Bitwise agreement at $\lambda = 2^{10}$, where no rounding error occurs at all, shows the implementation to be exactly scale-free. The deviations $\lesssim 10^{-13}$ at generic λ have the same slow amplification rate ($\sim 3.8 \times 10^{-4}/\text{step}$) as the twin perturbation and are dynamical amplification of initial rounding differences, not a breaking of scale symmetry. Together with Theorem 3, Axiom 0.5 holds both theoretically and numerically.

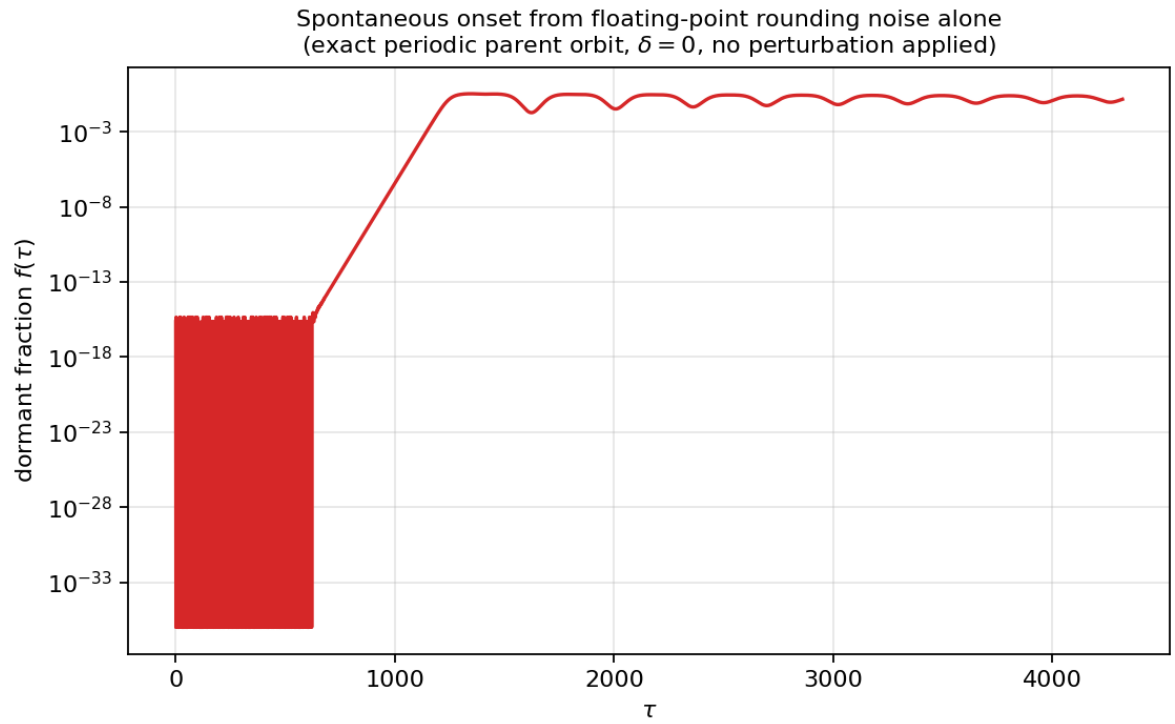


Figure 5: Collapse from rounding noise at $\delta = 0$

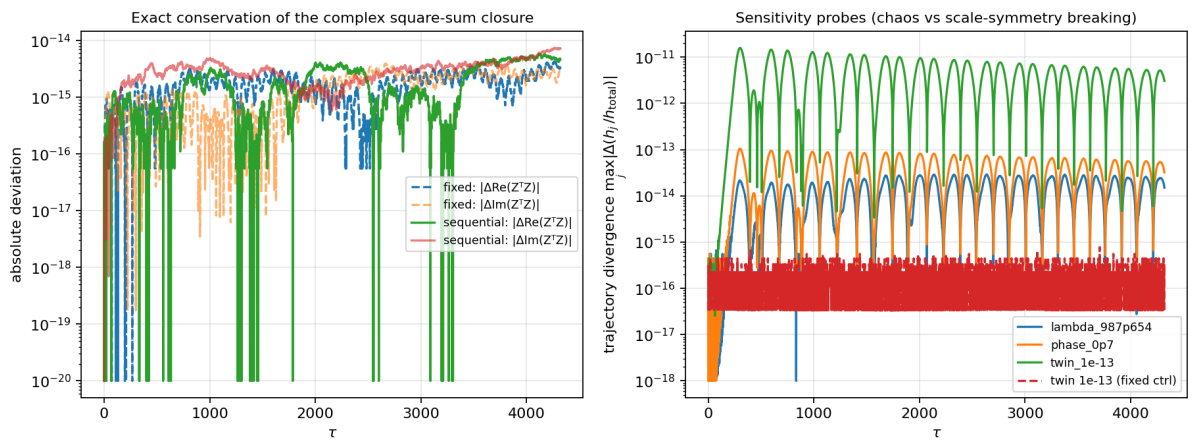


Figure 6: Closure conservation and sensitivity

Figure 6: Left — conservation error of the complex square-sum closure (machine precision for both dynamics). Right — sensitivity probes. Generic scale, common phase, and twin perturbation show comparable slow divergence; the fixed-generator reference does not diverge.

8.5 After onset: saturation, recurrence, non-establishment of a second plane ($N = 4$)

After saturation, f has a universal ceiling 0.335–0.338 and oscillates between 0.04 and 0.34. This is repeated splitting and recombination in a conservative system — phenomenology isomorphic to FPUT-type recurrence [7,8]. The origin of the ceiling value (structural or dynamical constant) is unresolved; no identification with any specific constant is made. Deciding this requires systematic measurement of the N and γ dependence.

Meanwhile the instantaneous generator remains single-plane throughout. σ_2 stays at 10^{-3} – 10^{-4} and never exceeds the relative threshold 0.05. Near the parent at $N = 4$, splitting as register reallocation occurs, but the establishment of a new rotation plane — splitting as an increase of the component count — does not.

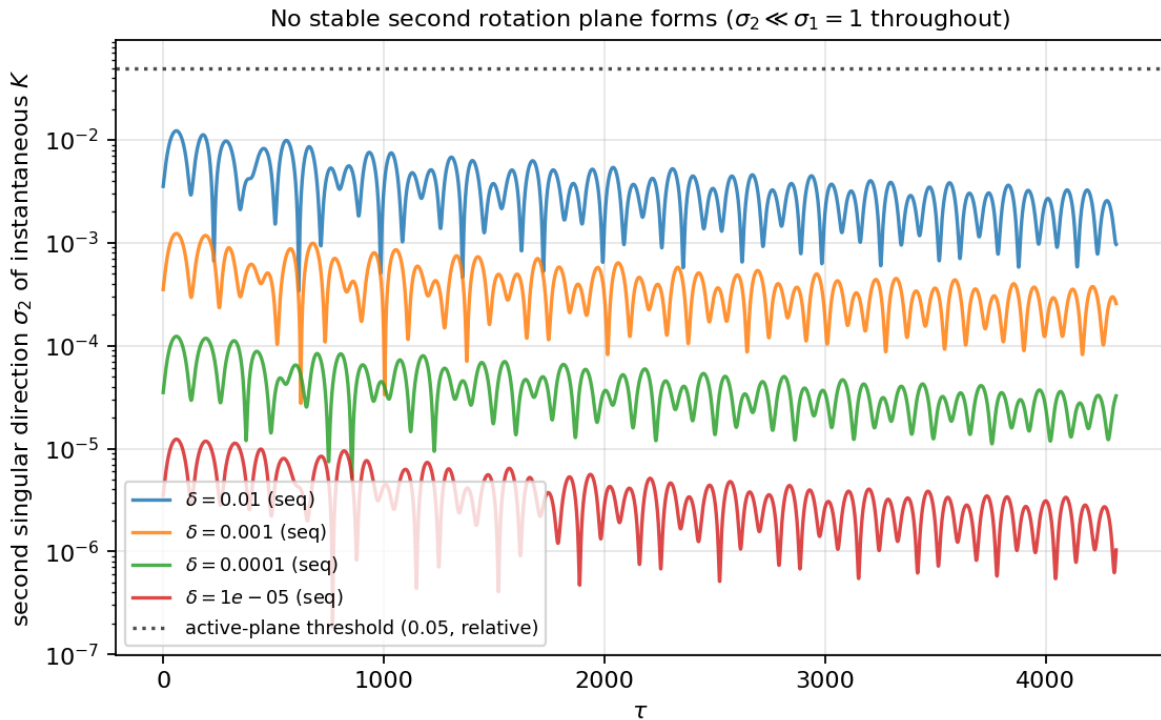


Figure 7: Non-establishment of the second singular direction

Figure 7: The second singular direction σ_2 of the instantaneous generator ($N = 4$, parent family). For all δ it stays below the activity threshold 0.05 — no second rotation plane establishes itself.

8.6 Threefold outcome classification (Experiment O4)

N	M	parent family	generic family
3	3	bounded mixing 6/6	bounded mixing 5/6, recurrence 1/6
4	6	bounded mixing 6/6 (σ_2 exp. decay)	bounded mixing 6/6
5	10	expansion 6/6	expansion 6/6
6	15	expansion 6/6	expansion 6/6

Across all trials the maximum deviation of closure conservation was 3.3×10^{-14} , and the initial zero closure was $|Z^T Z| \leq 2.3 \times 10^{-16}$ (except the $N = 3$ parent family, where no zero-closure seed exists and $O(\delta^2) = 10^{-6}$ is tolerated; Section 5.3). All three outcome classes were realized (recurrence in one trial only).

Expansion A new rotation plane establishes itself ($\sigma_2 > 0.05$ residence fraction 1) and the allocation spreads over all relational waves, converging to a stationary multi-plane state (parent-family representatives: $N = 5$: $f \rightarrow 0.55$, $\sigma_2 \approx 0.25$ stationary; $N = 6$: $f \rightarrow 0.8$, $\sigma_2 \approx 0.4$ stationary).

Bounded mixing The allocation spreads (in the parent family the participation ratio converges numerically to the equipartition value $PR = M$; median deviation $\lesssim 10^{-2}$), but no second plane establishes itself; a single plane wanders while mixing amplitudes only.

Recurrence Repeated deep returns toward the parent plane. In the zero-closure stratum ($c = 0$) it appeared in only one trial of the $N = 3$ generic family ($f_{\min} = 0.021$), but at $c \approx 0.55$, $N = 3$ shows it systematically in 3/3 (Section 8.7).

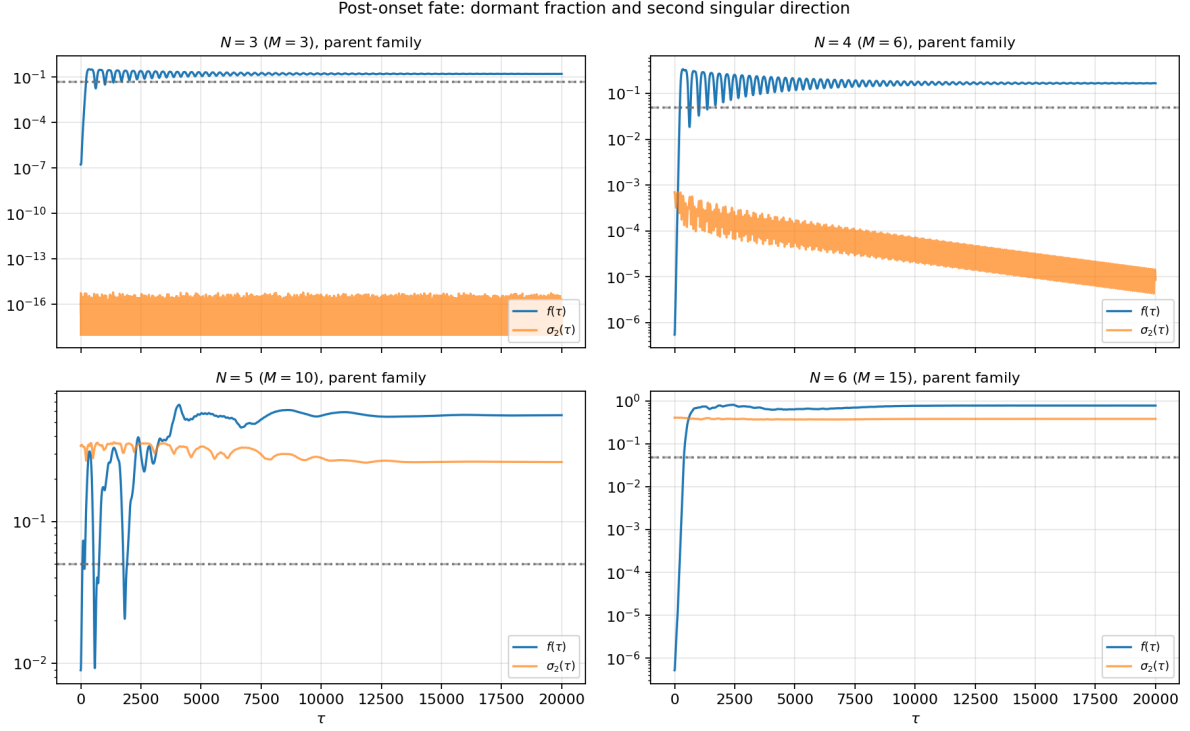


Figure 8: Fate curves of outcomes

Figure 8: Fate curves of the parent family ($N = 3, 4, 5, 6$). $N = 3$ has $\sigma_2 \equiv 0$ (expansion structurally impossible). At $N = 4$, σ_2 decays spontaneously and exponentially from 6×10^{-4} into the 10^{-5} range. For $N \geq 5$, σ_2 settles at a stationary value and expansion completes.

Figure 9: Classification map of all 48 trials (4 values of $N \times 2$ families $\times 6$ random seeds; horizontal: late recurrence depth, vertical: σ_2 active residence fraction). Expansion (top) and non-expansion (bottom) separate cleanly.

Figure 10: Late participation ratio. The parent family converges numerically to $PR = M$ (equipartition) for all N .

The structure of class selection is summarized in three points.

1. **N phase structure** At $N = 3$ expansion is structurally impossible: a skew-symmetric matrix with $M = 3$ has rank ≤ 2 , so no second rotation plane can exist ($\sigma_2 \equiv 0$, derived). At $N = 4$ no expansion arose from zero-closure states either (both families bounded mixing). For $N \geq 5$ expansion arose in both families and all random seeds. The boundary lies between $N = 4$ and $N = 5$ — though “for all initial conditions examined”: this is not a theorem over all initial states.
2. **Outcome stratification by the conserved ratio c** The controlled contrast (Experiment O5, Section 8.7) established that what separates expansion at $N = 4$ is not proximity to the parent but the exactly conserved ratio c . Restricting to $c = 0$ removes expansion for $N \leq 4$. That is, the closure condition of Axiom 1 (designation of the $c = 0$ stratum) is not only a conservation law but narrows the reachable end states of the dynamics. The mechanism is a connection problem.

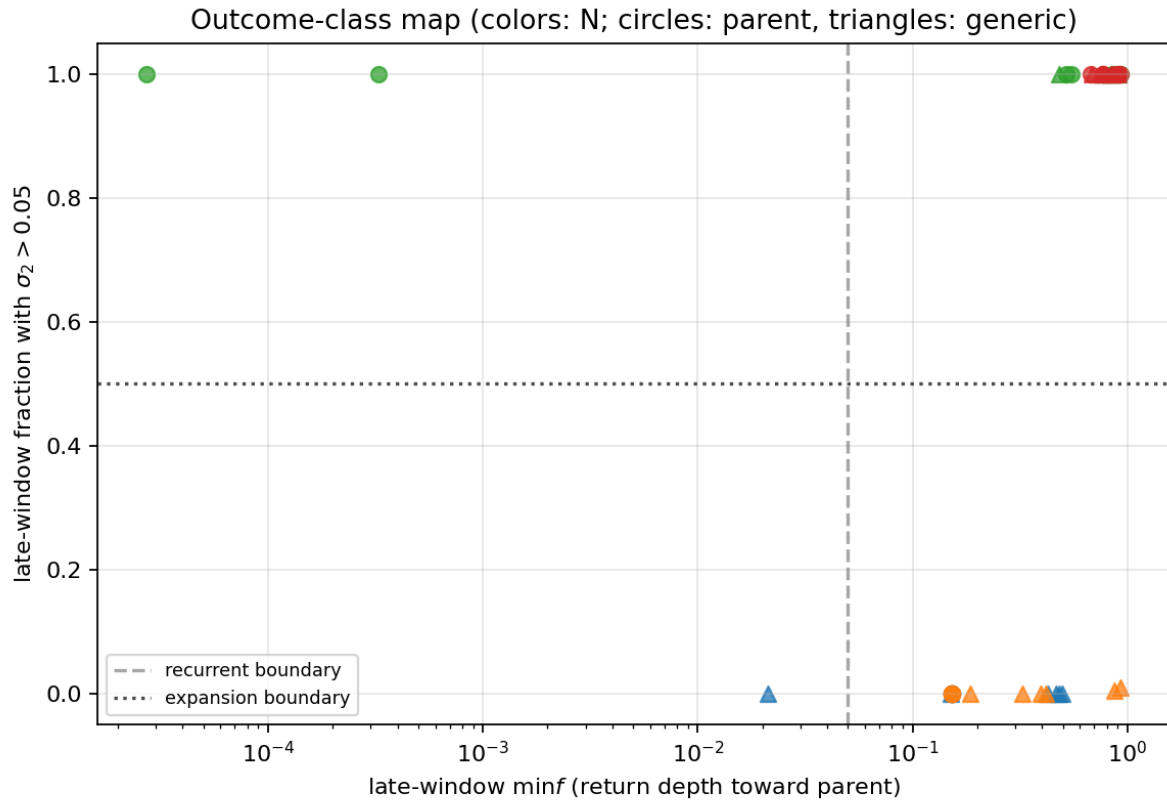


Figure 9: Outcome-class map

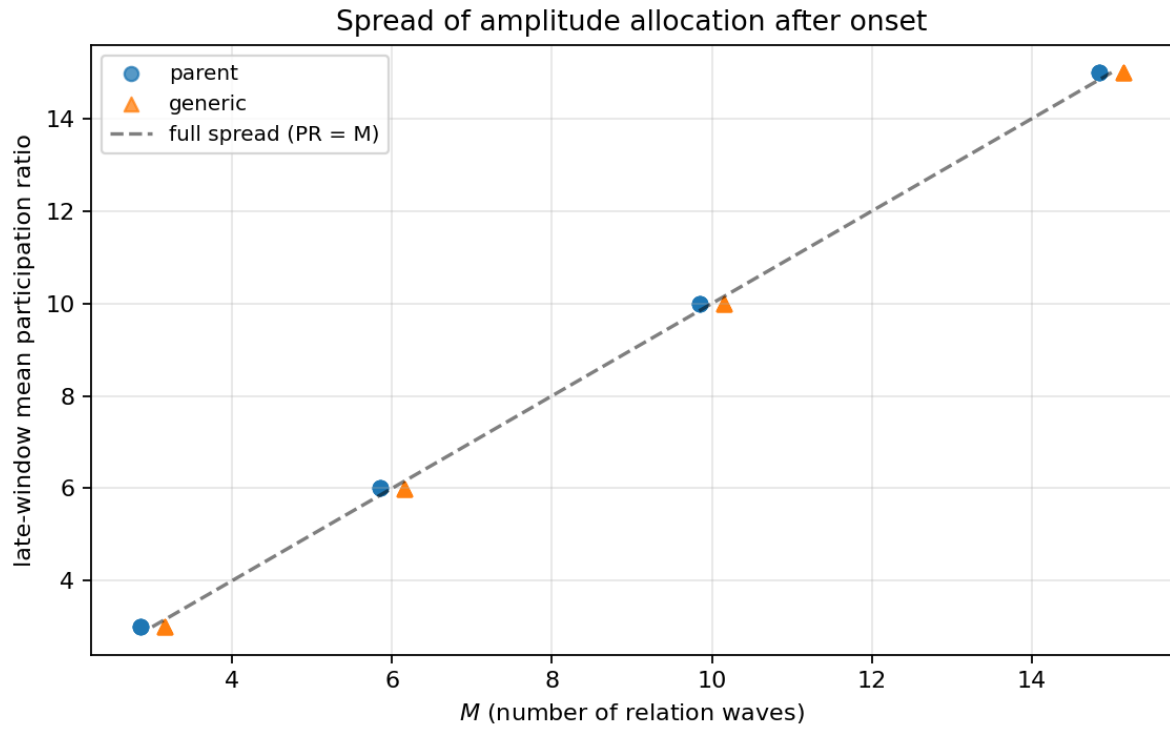


Figure 10: Spread of the allocation

3. **Observation of spontaneous refreezing** In the $N = 4$ parent family, σ_2 decayed exponentially without external intervention ($6 \times 10^{-4} \rightarrow 10^{-5}$ over 20000 steps), suggesting the system heads spontaneously toward a new single-plane self-consistent configuration — an effectively frozen state. However, $\sigma_2 \rightarrow 0$ only means the instantaneous generator approaches rank 2; if the orientation of the plane keeps moving, allocation transfer can continue. Confirming refreezing requires $\|K(\tau+1) - K(\tau)\|_F \rightarrow 0$ or vanishing drift of the principal-plane projector, which is done in the sequel.

8.7 Controlled contrast of the closure convention (Experiment O5): the conserved ratio c stratifies outcomes

N	genericZ0 ($c \approx 0$)	genericNZ ($c \approx 0.3-0.6$)	parentZ0 ($c \approx 0$)	parentNZ
3	bounded mixing 3/3	recurrence 3/3	bounded mixing 3/3	bounded mixing 3/3
4	bounded mixing 3/3	expansion 3/3	bounded mixing 3/3	bounded mixing 3/3
5	expansion 3/3	expansion 3/3	expansion 3/3	expansion 3/3
6	expansion 3/3	expansion 3/3	expansion 3/3	expansion 3/3

Maximum deviation of closure conservation across all 48 runs: 4.8×10^{-14} . Three readings are established.

1. **At small N (3, 4) the value of c changes the outcome class.** The controlled comparison between the generic families (everything identical, only c contrasted) separated systematically 3/3. For $N \geq 5$ it has no effect (all expansion)
2. **What acts is the magnitude of c .** The parentNZ family at $N = 3$ ($c \approx 10^{-6}$) fell into the same class as the $c = 0$ side, not as genericNZ ($c \approx 0.55$) — not a binary “zero versus nonzero” but the magnitude of the conserved ratio decides the stratum
3. **The old “near-parent versus generic” family contrast was confounded with c .** After separation by this controlled contrast, the variable separating the $N = 4$ outcome is not the character of the initial family but c

The recurrence class gained systematic support from this contrast: at $c \approx 0.55$, $N = 3$ recurs in 3/3; the recurrence that appeared in only one trial in Experiment O4 ($c = 0$) is a standard outcome in the nonzero- c strata.

Reservation: in the parentNZ family, by the nature of the seed construction, c came out unintentionally tiny ($\approx 10^{-16}$ for $N \geq 4$, 10^{-6} for $N = 3$), so the parent-side contrast is used only as evidence for point 2 (“tiny c behaves as $c = 0$ ”). The parentZ0 family at $N = 3$ degenerates (one-dimensional complement; Section 5.3).

8.8 Large- N onset runs (Experiment O6): body-count universality of the onset law and seed depth of 30 decades

	$N = 40$ ($M = 780$)	$N = 300$ ($M = 44850$)	$N = 1000$ ($M = 499500$)
$f(0)$	1.01×10^{-30}	1.40×10^{-30}	4.58×10^{-30}
parent residual	6.2×10^{-13}	9.3×10^{-13}	4.5×10^{-12}
initial growth rate (/step)	3.60×10^{-2}	2.49×10^{-2}	2.11×10^{-2}
steps per decade	64	93	109
$\tau_c(f \geq 0.05)$	2011	4844	7926
$ Z^\top Z $ max	1.4×10^{-15}	5.6×10^{-15}	2.7×10^{-14}

Three readings are established.

1. **The onset law holds at large N .** From seeds $f(0) \approx 10^{-30}$, 25 decades deeper than Experiment O1 ($N = 4$, $\delta \geq 10^{-5}$), the dormant fraction passed straight through more than 20 decades of constant-factor exponential amplification, then left the exponential regime and entered the saturation region where growth slows by orders. The seed-depth-independent exponential onset (Section 8.2) reproduces unchanged at body counts 40, 300, and 1000 (about half a million relational waves). The implication of Axiom 0.5 — only amplitude ratios act, and the arrival time of the amplification is linear in the seed’s decade count — is supported by direct measurement to this depth
2. **The rate depends only weakly on N .** The initial growth rate decreases monotonically from 3.8×10^{-2} at $N = 4$ through 3.6×10^{-2} at $N = 40$ and 2.5×10^{-2} at $N = 300$ to 2.1×10^{-2} at $N = 1000$, staying within the same order, with the decrease slowing at large N . Identifying the N -dependence law of the rate is a connection problem

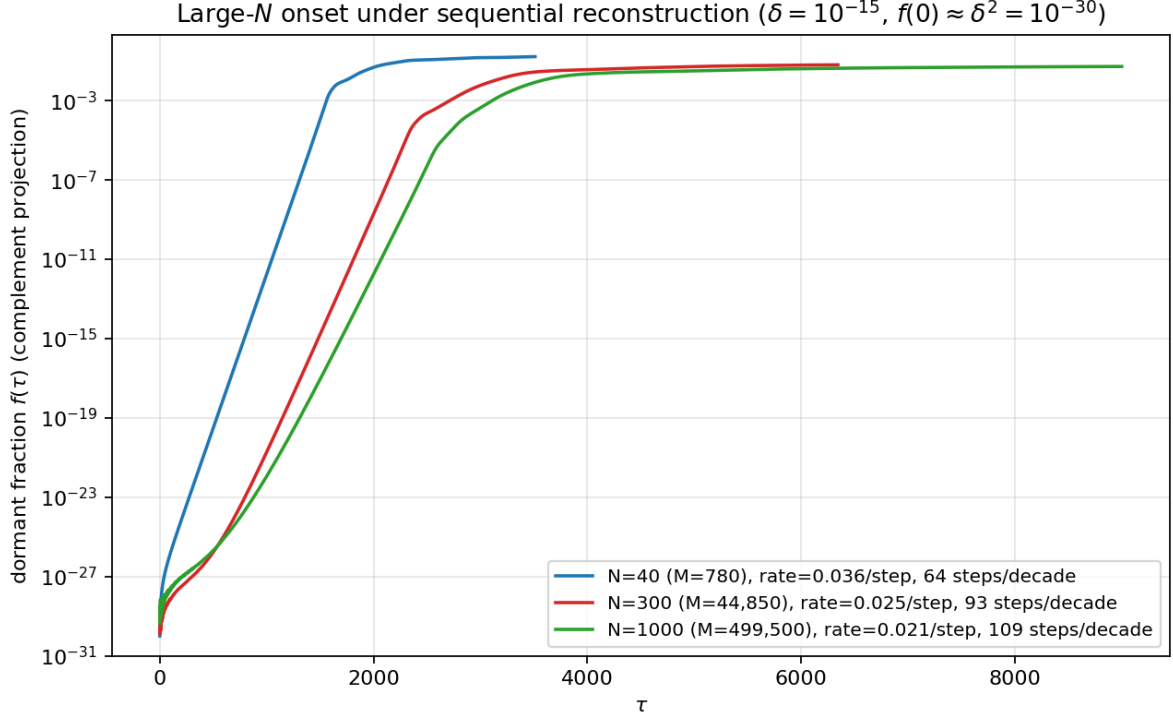


Figure 11: Large- N onset: constant-rate amplification from a seed depth of 30 decades

3. **The measurement floor is set by the parent’s construction accuracy.** The gentle initial rise from 10^{-30} to about 10^{-24} is a transient in which the construction residual of the parent eigenmode ($\sim 10^{-12}$, energy $\sim 10^{-24}$) leaks into the orthogonal complement; it is distinct from the subsequent exponential regime. The 20+ decades above the floor are the measured range of dynamical amplification

Reservation: one random seed per N ; no δ -series or seed statistics as in Experiment O1. Tracking extends to 1500 steps past crossing ($N = 1000$: 1074 steps past crossing due to the 9000-step cap), and the outcome classification of large N (the three classes of Section 8.6) is outside the scope of this experiment.

9. Discussion and Limitations

Relation to modulational instability The onset phenomenon of this paper shows phenomenology isomorphic to modulational instability [5,6]: exponential growth, seed-amplitude-independent rate, activation of discrete modes, saturation. The mechanism differs, however. NLS-type modulational instability arises from amplitude-dependent nonlinear coupling, whereas the generator of this system does not depend on amplitudes (phase-only coupling, zeroth-order homogeneous). It is not “a kind of MI” but “a phase-coupled instability with phenomenology isomorphic to MI.”

FPUT-type recurrence The post-saturation back-and-forth of splitting and recombination is isomorphic to recurrence in conservative systems [7,8]. The framework for judging recurrence versus thermalization (equipartition $PR = M$) follows [8]. The bounded-mixing class shows the intermediate aspect “recurring while the amplitudes head toward equipartition”; refining this classification is deferred to the sequel.

Determinism and predictability The collapse direction of the $\delta = 0$ run was decided by floating-point rounding. The update rule is fully deterministic, but the outcome depends sensitively on microscopic differences not contained in the macroscopic description [9]. Moreover, that the conserved global invariant c of the initial condition separates the outcome classes at small N shows that the value of a conserved quantity acts on outcome selection (Section 8.7).

Limitations (i) The operation of increasing the component count itself is an underdetermined freedom

the axioms grant from the start ($X_P^2 = \sum_k Y_k^2$ is always solvable over the complex numbers with a continuum of solutions) and requires no dynamical justification. That this paper selects no particular parameterization (no explicit splitting map) is not an omission: the physics — conservation, freezing, outcome selection — appears only in the readout layer (Section 1.2). “Splitting” in this paper is a statement of the readout layer: register reallocation and growth of the active-plane count within a state space of fixed dimension. (ii) The rank 2 property of the parent is a numerical insight, unproven. (iii) At larger N (some trials with $N \geq 5$) the parent fixed-point iteration did not converge and approximate parents were used. (iv) The classification thresholds ($\sigma_2 > 0.05$, $\min f < 0.05$, late window = second half) are preliminary; no sensitivity analysis was performed. (v) The origin of the saturation ceiling is unresolved. (vi) No theoretical formula for the growth rate 3.8×10^{-2} /step (Floquet analysis around the parent periodic orbit) has been carried out, and the quantitative resolution of the mismatch between the initial-fit rate and the crossing-derived effective rate (5.5×10^{-2}) — the non-single-exponential nature of the growth — also remains. (vii) The relative activity threshold 0.05 is a readout convention; its conventionality (the choice of how to count) is itself thematized in the sequel.

10. Connection Problems

1. **Stopping of splitting** (sequel, second question). The mechanism and rate of the spontaneous σ_2 decay in the $N = 4$ parent family. Identification of the destination of the decay (a new self-consistent configuration). Relation to external generator fixing (the operation that freezes all registers in the control experiment). The threshold of re-splitting.
 2. **Differentiation of conserved quantities** (sequel, third question). Classification of the relative components generated by splitting via exchange symmetry.
 3. Construction of an explicit splitting map (component-count increase, exact conservation of the complex square sum) and its relation to the growth of the active-plane count in this paper.
 4. Floquet analysis of the parent periodic orbit and comparison with the measured growth rate.
 5. The mechanism of outcome stratification by the conserved ratio c — why expansion is suppressed for $N \leq 4$ in the $c = 0$ stratum; measurement of the stratum boundary (the critical c at which classes switch).
 6. Systematic measurement of the N and γ dependence of the saturation ceiling.
 7. Proof of the parent’s rank 2 property; the existence problem of self-consistent parents at larger N .
 8. Statistics of selective freezing (distribution of freezing times and frozen allocations).
-

11. Reproducibility

The numerical experiments of this paper are based on the following programs and outputs.

Programs

- `dimension_generation_structure/spontaneous_splitting_preliminary_v1/run_spontaneous_splitting_p`
(Experiments N, O1, scale verification, $\delta = 0$ verification)
- `dimension_generation_structure/spontaneous_splitting_preliminary_v1/run_outcome_classification_p`
(Experiment O4)
- `dimension_generation_structure/spontaneous_splitting_preliminary_v1/run_closure_contrast_v1.py`
(Experiment O5)
- `dimension_generation_structure/spontaneous_splitting_preliminary_v1/run_spontaneous_splitting_la`
(Experiment O6)
- `dimension_generation_structure/spontaneous_splitting_preliminary_v1/run_n_scaling_lowrank_v1.py`
(low-rank vertex-decomposition engine; dense-matrix agreement verification)
- `dimension_generation_structure/spontaneous_splitting_preliminary_v1/make_largeN_figure_v1.py`
(Experiment O6 figure)
- `dimension_generation_structure/spontaneous_splitting_preliminary_v1/make_paper_supplement_figure`
(supplementary figures)

Main outputs

- `spontaneous_splitting_result_v1/summary_v1.json`, `dormant_fraction_curves_v1.csv`, 7 figures
 - `outcome_classification_result_v1/summary_v1.json`, 3 figures
 - `closure_contrast_result_v1/summary_v1.json` (Experiment O5)
 - `largeN_splitting_result_v1/fcurve_*.csv`, `summary_*.json`, 1 figure (Experiment O6)
-

12. Final Conclusions

Dynamics preserving the closure sphere is exhausted by $\dot{X} = K(X)X$ (K skew-symmetric); the fixed generator is a special case. Merely unfreezing the sine generator — which was defined as a function of phase differences from the start — without introducing any new coupling rule and without external forcing, the parent state decays exponentially at an intrinsic rate independent of the seed amplitude, while the real and imaginary parts of the complex square-sum closure are conserved to machine precision and the allocation moves spontaneously into dormant directions. Even the exact periodic orbit at $\delta = 0$ collapses through the amplification of the nonzero numerical perturbation of rounding. Just outside the boundary drawn by the no-go of Paper 4, spontaneous splitting begins.

Existence opens without limit; only the readout saturates. The rank bound and three-direction saturation of Paper 3, the register freezing of Paper 4, and the outcome classification of this paper are all statements of readout saturation, not statements of the finiteness of existence.

Splitting can begin. But the outcome is not a single road.

Three classes are real — expansion (establishment of a new plane, equipartition), bounded mixing (wandering of a single plane), recurrence (repeated returns toward the parent) — and the selection is decided by the N phase structure ($N \leq 4$ no expansion, $N \geq 5$ expansion in all trials) and the conserved ratio c .

Scale anonymity (Axiom 0.5) was introduced as the formation rule running through this entire construction. The unfrozen generator satisfies it without adjustment as a zeroth-order homogeneous function; the readouts and thresholds are all ratio quantities; and numerically all ratio observables agreed bitwise under a 2^{10} -fold scale change. The zero square sum of Axiom 1 is the unique quadratic closure compatible with this axiom.

The spontaneous exponential decay of σ_2 observed in the $N = 4$ parent family suggests that splitting can stop without an external freezing operation. Can splitting begin? — the answer of this paper is yes. Can it stop? — that is the question of the sequel.

References

Self-citations

1. Noriaki Kihara, “Basic Axiom System of the Anonymous Equal-Amplitude Composite Wave Model v7,” Zenodo, 2026. Version DOI: 10.5281/zenodo.21469133, Concept DOI: 10.5281/zenodo.21315735.
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